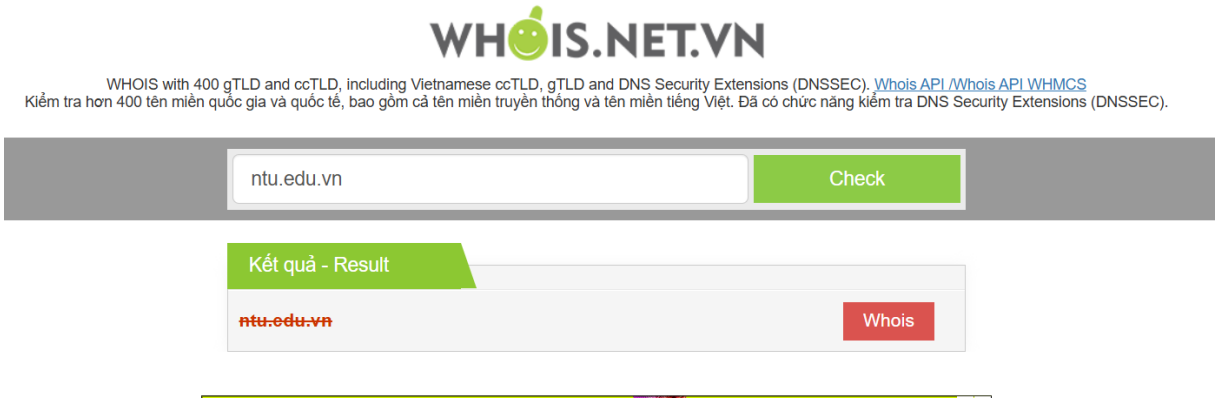
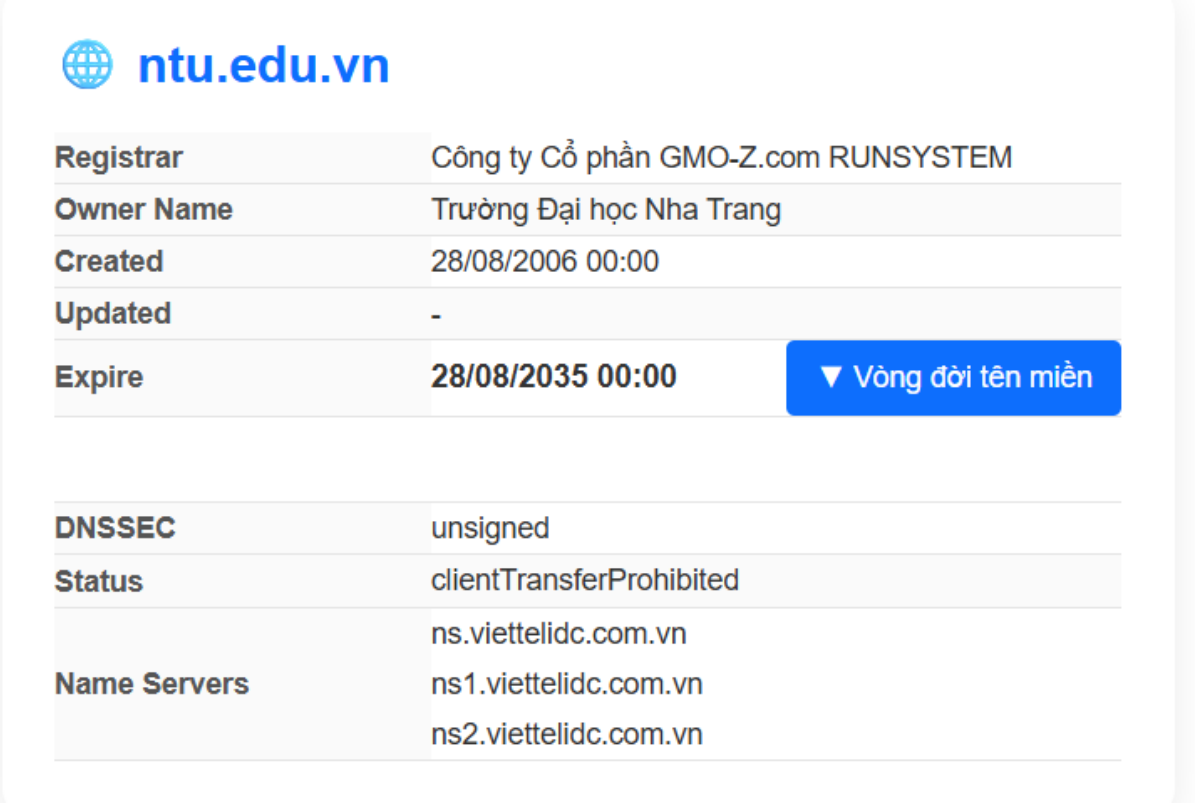
Sử dụng các công cụ dò thám thông tin, hãy dò thám thông tin bảo mật của mạng Trường đại học nha trang thông qua URL https://ntu.edu.vn/

1. Thu thập thông tin định danh tên miền trên https://whois.net.vn/



Kết quả nhận được:



2. Phân tích hạ tầng DNS trên Ubuntu

2.1 Xác định địa chỉ IP máy chủ

```
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ nslookup ntu.edu.vn
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   ntu.edu.vn
Address: 118.69.120.28
```

2.2 Xác định máy chủ phân giải tên miền

```
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ dig ntu.edu.vn NS +short
ns1.viettelidc.com.vn.
ns2.viettelidc.com.vn.
ns.viettelidc.com.vn.
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$
```

2.3 Xác định hệ thống máy chủ thư điện tử (Mail Server)

```
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ dig ntu.edu.vn MX +short
40 ASPMX2.GOOGLEMAIL.COM.
50 ASPMX3.GOOGLEMAIL.COM.
30 ALT2.ASPMX.L.GOOGLE.COM.
20 ALT1.ASPMX.L.GOOGLE.COM.
10 ASPMX.L.GOOGLE.COM.
```

2.4 Đánh giá cấu hình chống giả mạo Email (SPF & DMARC)

```
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ dig ntu.edu.vn TXT +short
"_globalsign-domain-verification=bJlB6dafsOZO5UmjIKTM6djIsTS9qKsZS2KZT9WfyK"
"v=spf1 include:_spf.google.com ~all"
"_globalsign-domain-verification=zlC0CMbJj7Xwfbr8FGvte_C5tf40V3xZhJWqmphoMy"
"_globalsign-domain-verification=eHeShhBviqh-RijMCnTUSKJOh_HCD7Yl5R4d9Fmu_p"
"zalo-platform-site-verification=OeM5BBsqDt1vWvrnlu4MHaIXbcoetaqtDJS"
"google-site-verification=ie7oXhpBA7RTAQMnrpjCCzTerCNRvgSEdO-Wkc6cATM"
"google-site-verification=6R9rbR6X1Ed7CSwO_D9q7xiEBvqTSSXbg-tE_WftgyU"
"google-site-verification=v2X8gdM6TfvTraZbM89n32s1z4kcrqqGDmnUhEhJ9Gk"
"_globalsign-domain-verification=oYfbXw44z5Bb-R5jWVxASa66WIyQpx1MlTawOe1xTc"
"MS=E5D7A62F3835113BD54DE6E0429C61E8CBA4A291"
"_globalsign-domain-verification=zs-fGawDmlHz2QJi71SHBK9GRxkkvqbDYCBLJ3iRY0"
```

3. Phân tích tiêu đề phản hồi Web (HTTP Security Headers)

```
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ curl -I https://ntu.edu.vn
HTTP/2 200
cache-control: no-cache,public, max-age=604800
pragma: no-cache
content-length: 674272
content-type: text/html; charset=utf-8
expires: -1
vary: Accept-Encoding
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-aspnetmvc-version: 5.3
x-ua-compatible: IE=edge
set-cookie: dnn_IsMobile=False; path=/; HttpOnly
set-cookie: .ASPXANONYMOUS=jROgo4Sj6c2G1BAXcEQhFzsqM-WmHTFnT58kJ8PrpRa1xZJ016plOOtTmeWOOSsda8Elz9S7YZfrq_6SVJ5niBVFTvYHtz0FYbDog7AyIr3vmcoy0; expires=Fri, 08-Jan-2027 04:
04:18 GMT; path=/; HttpOnly
set-cookie: language=vi-VN; path=/; HttpOnly
set-cookie: __RequestVerificationToken=3MV7UrgGa_N0nBAnXbM_JnRI2_HKhZWHYufQwKeWEqHPHe47vCVPgdaz1RJw8jciIv3OmQ2; path=/; HttpOnly
x-xss-protection: 1; mode=block
x-frame-options: SAMEORIGIN
x-content-type-options: nosniff
strict-transport-security: max-age=31536000; includeSubDomains; preload
x-permitted-cross-domain-policies: none
permissions-policy: unload=*
content-security-policy: frame-ancestors 'self'
expect-ct: max-age=7776000, enforce, report-uri= ;
date: Fri, 30 Oct 2026 17:24:17 GMT
```

4. Nhận diện nền tảng công nghệ (Web Fingerprinting)

```
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ whatweb https://ntu.edu.vn
https://ntu.edu.vn [200 OK] ASP_NET[MVC5.3, 5.3, 5.3, 5.3, 5.3, 5.3, 5.3, 5.3, 5.3, 5.3, 5.3, 5.3, 5.3,ViewState Encrypted], Bootstrap, Cookies[.ASPXANONYMOUS,__Requ
estVerificationToken,dnn_IsMobile,language], Country[VIET NAM][VN], DotNetNuke[Admin,MVC], Email[dhnt@ntu.edu.vn], Frame, HTML5, HttpOnly[.ASPXANONYMOUS,__RequestVerifica
tionToken,dnn_IsMobile,language], IP[118.69.120.28], JQuery, Open-Graph-Protocol[Website], Script[javascript,text/javascript], Strict-Transport-Security[max-age=31536000;
 includeSubDomains; preload], Title[Trường Đại học Nha Trang - Nha Trang University][Title element contains newline(s)!], UncommonHeaders[x-aspnetmvc-version,x-content-ty
pe-options,x-permitted-cross-domain-policies,permissions-policy,content-security-policy,expect-ct], X-Frame-Options[SAMEORIGIN], X-UA-Compatible[IE=edge], X-XSS-Protectio
n[1; mode=block], YouTube
```

5. Kiểm tra dịch vụ mạng trên cổng Web (Service Scan)

```
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ nmap -sV -p 80,443 -Pn ntu.edu.vn
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-10-31 00:29 +07
Nmap scan report for ntu.edu.vn (118.69.120.28)
Host is up (0.028s latency).

PORT    STATE SERVICE  VERSION
80/tcp  open  http     Microsoft IIS httpd 10.0
443/tcp open  ssl/http Microsoft IIS httpd 10.0
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 29.27 seconds
```

6. Kiểm tra chi tiết chứng chỉ số SSL/TLS bằng openssl

```
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ openssl s_client -connect ntu.edu.vn:443 -servername ntu.edu.vn
CONNECTED(00000003)
depth=2 OU = GlobalSign Root CA - R6, O = GlobalSign, CN = GlobalSign
verify return:1
depth=1 C = BE, O = GlobalSign nv-sa, CN = GlobalSign GCC R6 AlphaSSL CA 2025
verify return:1
depth=0 CN = *.ntu.edu.vn
verify return:1
---
Certificate chain
 0 s:CN = *.ntu.edu.vn
   i:C = BE, O = GlobalSign nv-sa, CN = GlobalSign GCC R6 AlphaSSL CA 2025
   a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
   v:NotBefore: Oct 30 03:04:58 2025 GMT; NotAfter: Dec  1 03:04:57 2026 GMT
 1 s:C = BE, O = GlobalSign nv-sa, CN = GlobalSign GCC R6 AlphaSSL CA 2025
   i:OU = GlobalSign Root CA - R6, O = GlobalSign, CN = GlobalSign
   a:PKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
   v:NotBefore: May 21 02:36:52 2025 GMT; NotAfter: May 21 00:00:00 2027 GMT
---
```

```
Server certificate
-----BEGIN CERTIFICATE-----
MIIGVzCCBT+gAwIBAgIMCE6U4pi8+DXOLYRCMA0GCSqGSIb3DQEBCwUAMFUxCzAJ
BgNVBAYTAkJFMRkwFwYDVQQKExBHbG9iYWxTaWduIG52LXNhMSswKQYDVQQDEyJH
bG9iYWxTaWduIEdDQyBSNiBBbHBoYVNTTCBDQSAyMDI1MB4XDTI1MTAzMDAzMDQ1
OFoXDTI2MTIwMTAzMDQ1N1owFzEVMBMGA1UEAwwMKi5udHUuZWR1LnZuMIIBIjAN
BgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAuDs/epM6xq9cicUIKKxNtooM6AOh
EqBZg5R2afpsrnXsM/hGhnAy0hT09iKTOk5REK0R4TiDkNk0RbQyBf0aGuM06w0I
juzgjN/49usiK4nMzJy7JHcuc2VqxRASniyJ8cQckBdscnP+C9U1hKbE7jNp7RU0
NYN5X9DLb54uowEviuECHZaNwBB022f7Xdd/v+p/2vuaVV3EJ8LC39S8mSpCLSzw
wu36ehDCOMQcZrp/auuQFFlie2B/TLR38ZBHF3KE0600y/6n2ONsrUTm+yrBsy03
TYylJcPK5GkZ1MIGBx0+SUHDO38rytKxGT51pyoW91/BJe8EyUQkNiDDewIDAQAB
o4IDYzCCA18wDgYDVR0PAQH/BAQDAgWgMAwGA1UdEwEB/wQCMAAwgZkGCCsGAQUF
BwEBBIGMMIGJMEkGCCsGAQUFBzAChj1odHRwOi8vc2VjdXJlLmdsb2JhbHNpZ24u
Y29tL2NhY2VydC9nc2djY3I2YWxwaGFzc2xjYTIwMjUuY3J0MDwGCCsGAQUFBzAB
hjBodHRwOi8vb2NzcC5nbG9iYWxzaWduLmNvbS9nc2djY3I2YWxwaGFzc2xjYTIw
MjUwVwYDVR0gBFAwTjAIBgZngQwBAgEwQgYKKwYBBAGgMgoBAzA0MDIGCCsGAQUF
BwIBFiZodHRwczovL3d3dy5nbG9iYWxzaWduLmNvbS9yZXBvc2l0b3J5LzBEBgNV
HR8EPTA7MDmgN6A1hjNodHRwOi8vY3JsLmdsb2JhbHNpZ24uY29tL2dzZ2NjcjZh
bHBoYXNzbGNhMjAyNS5jcmwwIwYDVR0RBBwwGoIMKi5udHUuZWR1LnZuggpudHUu
ZWR1LnZuMB0GA1UdJQQWMBQGCCsGAQUFBwMBBggrBgEFBQcDAjAfBgNVHSMEGDAW
gBTFtJOPbyvcHki/txAwhc7RsrtILTAdBgNVHQ4EFgQUXiirFcRLgF04j6gJzOc+
R6kWxuYwggF+BgorBgEEAdZ5AgQCBIIBbgSCAWoBaAB2AMs49xWJfIShRF9bwd37
yW7ymlnNRwppBYWwyxTDFFjnAAABmjMTazAAAAQDAEcwRQIhAPXZ9veM3xmLtyZh
PVGXt6IqZQz0eHDdhmoWq27kEcR2AiBzrwZpKwZT/oNojn8knxI36Wy22oAxJJvd
VCWK7OC4AAB2ANgJVTuUT3r/yBYZb5RPhauw+Pxeh1UmDxXRLnK7RUsUAAABmjMT
bF4AAAQDAEcwRQIgM5ozNyUENJwEK/0ptV27exK1f2kyyBkx/HBKSRTMGwUCIQCo
l1aXvd6pE1LFt65ScJE8zwXBvnVkXAKVNjsDXD+22gB2AMIxfldFGaNF7n843rKQ
QevHwiFaIr9/1bWtdprZDlLNAAABmjMTbLEAAAQDAEcwRQIhAOlsDJ9850i5WROp
7x1z/9CMRoQEJR8tPyvDwlXXT1S2AiBYJmTgpVIjeEJ3VCAm2Xg1qPtIVIsg818E
fKzr/QrgBjANBgkqhkiG9w0BAQsFAAOCAQEAJ6Y/CwdYMI+2SS1vIL/7a/PdGgdP
hnlLcWfzkeChoy7U5qD3ZMG45saAKFxi6SSJVuyfl9YZNsz/FImIRnVyfcnJNS9A
G6cSc8qkvTDgJLlWiJZa4Z0dhS66nhoTCEZ3UWUZBWtAmrncjhTDZI/hxld3NEGE
eBK0eQ9Vh8KWk4Qvi0hL+906XFkd/QUa8i+HCfGOdm9FZpG4lZT5HHgATfDIm3Ts
0tiKoVjcGKTOIHRFvCe+ot02MwkcjF64WeLyIluy7u0+MOM9j5IApKYaq/QmkR+8
zonZHk9OlQ/cbR4jchPHKUEZwhLwRlfLuv0tC4SUE3ZSTrmfEui0zV2U+A==
-----END CERTIFICATE-----
```

```
subject=CN = *.ntu.edu.vn
issuer=C = BE, O = GlobalSign nv-sa, CN = GlobalSign GCC R6 AlphaSSL CA 2025
---
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: RSA-PSS
Server Temp Key: X25519, 253 bits
---
SSL handshake has read 3547 bytes and written 392 bytes
Verification: OK
---
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Server public key is 2048 bit
Secure Renegotiation IS NOT supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 0 (ok)
---
```

```
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol  : TLSv1.3
    Cipher    : TLS_AES_256_GCM_SHA384
    Session-ID: 07DE93D47A1A70FA0C91C78F39512B05388706869DED4FBA18644BA67C1206EE
    Session-ID-ctx:
    Resumption PSK: 803E2F1A10DF0401FB2F705B233ABEDA4F453FF1B39B01F9DF2425CDE1BBC1B36D625059CFA056B4D66234F5B803E463
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 36000 (seconds)
    TLS session ticket:
    0000 - 9e 2e 00 00 a5 21 c2 f4-50 20 c5 f3 f3 e7 a6 f6   .....!..P ......
    0010 - 4f 8b 54 55 82 e3 da f1-f5 5c d8 89 d9 b6 7b 0a   O.TU.....\....{.

    Start Time: 1793381552
    Timeout   : 7200 (sec)
    Verify return code: 0 (ok)
    Extended master secret: no
    Max Early Data: 0
---
```

7. Kiểm tra chính sách DMARC cho hệ thống Email

```
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ dig _dmarc.ntu.edu.vn TXT +short
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$
```

Hệ thống chưa thiết lập DMARC (Domain-based Message Authentication, Reporting, and Conformance)

8. Kiểm tra tệp tin chỉ mục tìm kiếm (robots.txt)

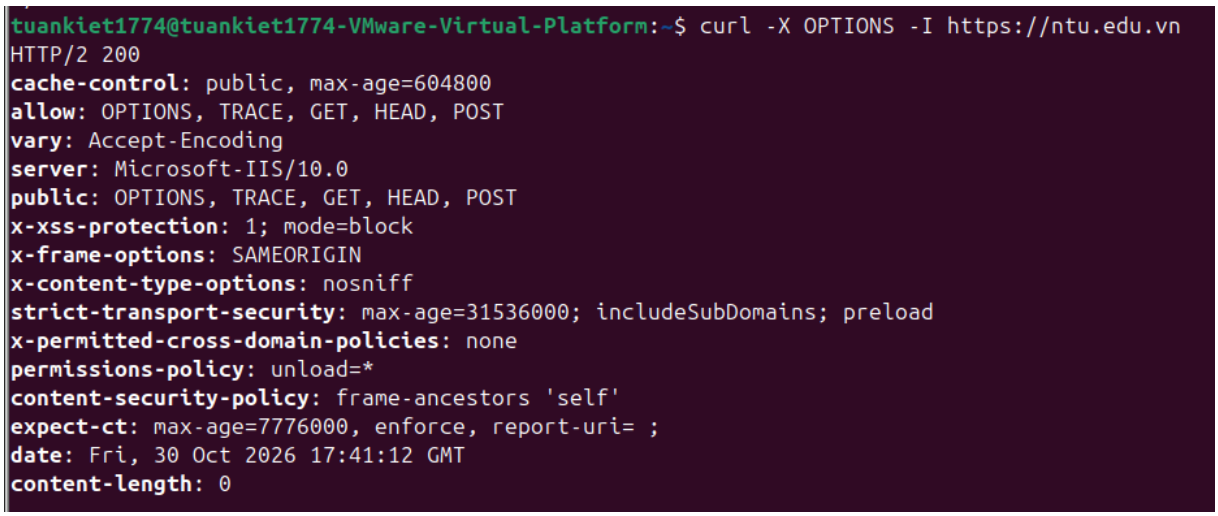
```
tuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ curl -s https://ntu.edu.vn/robots.txt
# Begin robots.txt file
#/-----------------------------------------------\
#| In single portal/domain situations, uncomment the sitmap line and enter domain name
#\-----------------------------------------------/
Sitemap: http://www.ntu.edu.vn/sitemap.aspx

User-agent: *
Disallow: /*/ctl/               # Googlebot permits *
Disallow: /admin/
Disallow: /App_Browsers/
Disallow: /App_Code/
Disallow: /App_Data/
Disallow: /App_GlobalResources/
Disallow: /bin/
Disallow: /Components/
Disallow: /Config/
Disallow: /contest/
Disallow: /controls/
Disallow: /Documentation/
Disallow: /HttpModules/
Disallow: /Install/
Disallow: /Providers/
Disallow: /Activity-Feed/userId/        # Do not index user profiles
Disallow: /Files/
Disallow: /fileman/
Disallow: /Scripts/
# End of robots.txt filetuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$
```

9. Kiểm tra tệp liên hệ bảo mật chuẩn RFC 9116 (security.txt)

```
# End of robots.txt filetuankiet1774@tuankiet1774-VMware-Virtual-Platform:~$ curl -s -L https://ntu.edu.vn/.well-known/security.txt
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1"/>
<title>404 - File or directory not found.</title>
<style type="text/css">
<!--
body{margin:0;font-size:.7em;font-family:Verdana, Arial, Helvetica, sans-serif;background:#EEEEEE;}
fieldset{padding:0 15px 10px 15px;}
h1{font-size:2.4em;margin:0;color:#FFF;}
h2{font-size:1.7em;margin:0;color:#CC0000;}
h3{font-size:1.2em;margin:10px 0 0 0;color:#000000;}
#header{width:96%;margin:0 0 0 0;padding:6px 2% 6px 2%;font-family:"trebuchet MS", Verdana, sans-serif;color:#FFF;
background-color:#555555;}
#content{margin:0 0 0 2%;position:relative;}
.content-container{background:#FFF;width:96%;margin-top:8px;padding:10px;position:relative;}
-->
</style>
</head>
<body>
<div id="header"><h1>Server Error</h1></div>
<div id="content">
 <div class="content-container"><fieldset>
  <h2>404 - File or directory not found.</h2>
  <h3>The resource you are looking for might have been removed, had its name changed, or is temporarily unavailable.</h3>
 </fieldset></div>
</div>
</body>
</html>
```

10. Kiểm tra các phương thức HTTP cho phép (HTTP Methods)



allow: OPTIONS, TRACE, GET, HEAD, POST

➔ Máy chủ đang bật phương thức TRACE ➔ Đây là một cấu hình sai sót về mặt bảo mật (Security Misconfiguration)

Phương thức TRACE được thiết kế để kiểm tra đường truyền (phản hồi lại nguyên văn yêu cầu của client), nhưng có thể bị lợi dụng trong tấn công Cross-Site Tracing (XST) để đánh cắp Cookie phiên (ngay cả khi Cookie có gắn cờ HttpOnly).